

CO 1 - ASSESSMENT TOOL -2

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Subject Code : CSA1003
Subject Name : Software Engineering

Hotel Reservation and Room Management Platform with Dynamic Pricing

Agile Sprint Planning & User Story Workshop

1. Project Vision and Stakeholders

Project Vision

To develop a secure, user-friendly, and scalable hotel reservation platform that enables customers to search, book, and pay for hotel rooms online while allowing hotel administrators to manage room availability, reservations, and dynamic pricing efficiently.

Objectives

- Enable online room reservations.
- Automate room management.
- Implement dynamic pricing based on demand and occupancy.
- Reduce manual errors and double bookings.
- Improve customer satisfaction.
- Increase hotel revenue.

Stakeholders

Stakeholder	Role
Customers	Search and book hotel rooms
Hotel Administrator	Manage rooms, pricing, bookings, reports
Hotel Staff	Check-in, check-out, housekeeping
Payment Gateway	Process online payments securely
System Administrator	Maintain system, security, backups

2. Epics and User Stories

Epic 1: User Management

User Story 1

As a customer, I want to register an account so that I can book hotel rooms online.

User Story 2

As a customer, I want to log in securely so that I can access my bookings.

Epic 2: Room Booking

User Story 3

As a customer, I want to search available rooms so that I can choose the best room.

User Story 4

As a customer, I want to reserve a room online so that my booking is confirmed instantly.

Epic 3: Dynamic Pricing

User Story 5

As a hotel administrator, I want room prices to change automatically based on demand so that revenue is maximized.

Epic 4: Payment

User Story 6

As a customer, I want to pay online securely so that my reservation is completed immediately.

Epic 5: Reports

User Story 7

As a hotel administrator, I want booking and revenue reports so that I can monitor business performance.

3. Acceptance Criteria

User Story 1

Given a new customer

When valid registration details are entered
Then the account should be created successfully.

User Story 3

Given rooms are available
When the customer searches using check-in and check-out dates
Then available rooms should be displayed.

User Story 4

Given a room is available
When the customer confirms booking
Then the reservation should be saved and confirmation displayed.

User Story 5

Given occupancy exceeds the configured threshold
When the pricing engine updates rates
Then room prices should increase automatically according to the pricing rules.

User Story 6

Given booking details are confirmed
When payment is successful
Then the booking status should become "Confirmed."

4. Product Backlog (Prioritized)

Priority Product Backlog Item

High	User Registration
High	User Login
High	Search Rooms
High	Room Booking
High	Online Payment
Medium	Dynamic Pricing
Medium	Booking Cancellation

5. Sprint Backlog (Sprint 1)

Sprint Duration (2 Weeks)

Selected User Stories

- User Registration
- User Login
- Search Available Rooms
- Book Room
- Online Payment

Development Tasks

- Design database
- Create login page
- Build room search module
- Develop booking functionality

6. Story Point Estimation

User Story	Story Points	Justification
Registration	3	Simple form with validation
Login	2	Authentication module
Search Rooms	5	Database queries and filters
Room Booking	8	Reservation logic and validation
Online Payment	8	Third-party payment integration
Dynamic Pricing	13	Complex pricing algorithms
Reports	5	Report generation and analytics

7. Sprint Goal and Expected Deliverables

Sprint Goal

Develop the core hotel reservation features that allow customers to create accounts, log in, search for available rooms, book rooms, and make secure online payments.

Expected Deliverables

- User Registration Module
- Login Module
- Room Search Module
- Reservation Module
- Online Payment Integration